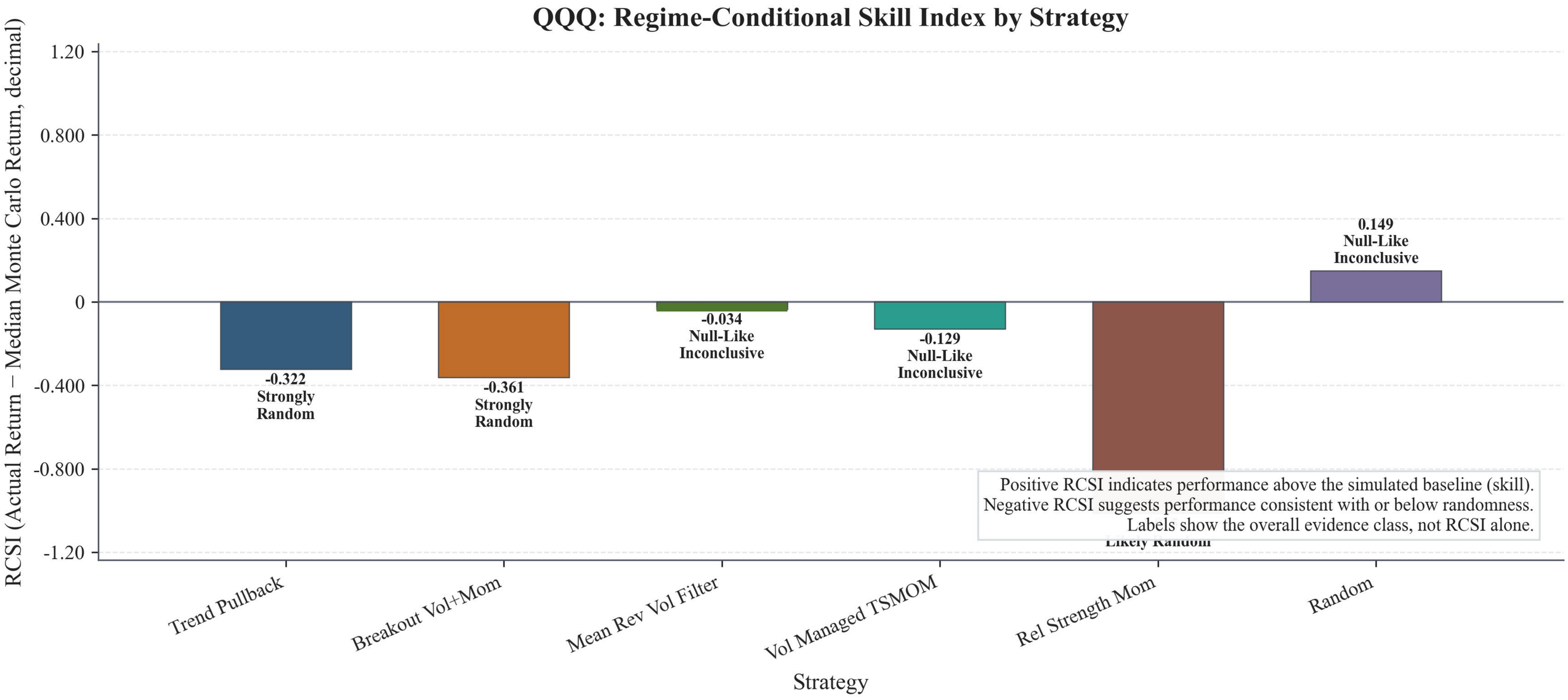
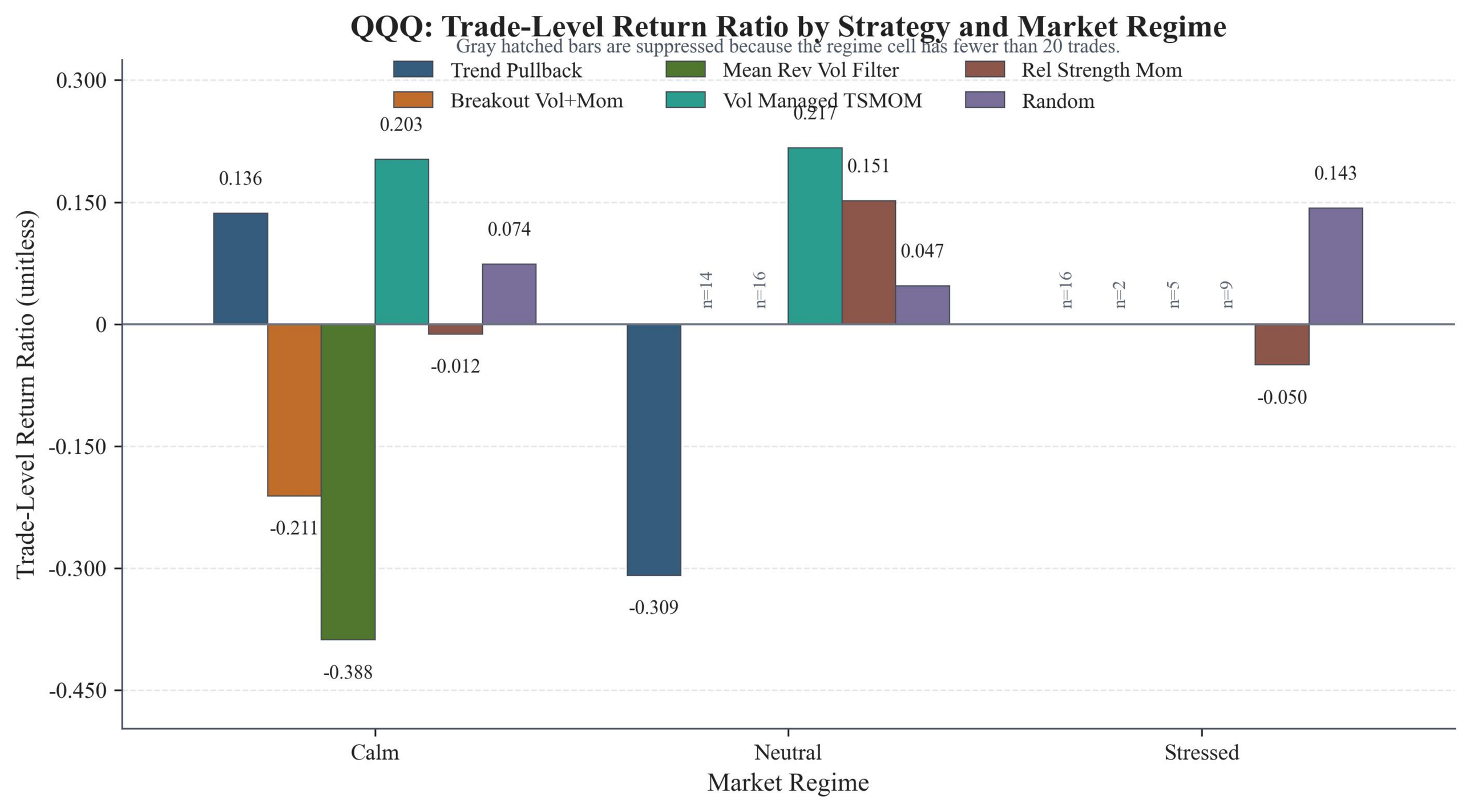
QQQ: Strategy Evidence and Confidence Overview

Evidence labels use scale-free statistics: p-value, percentile, normalized RCSI, robustness, and stability.

Strategy	Evidence	Verdict	Confidence	p-value	Prom.	Pctile	RCSI_z
Trend Pullback	Strongly Random	Luck	High	0.878	0.06	12.2	-1.07
Breakout Vol+Mom	Strongly Random	Luck	High	0.925	0.03	7.5	-1.29
Mean Rev Vol Filter	Null-Like / Inconclusive	Inconclusive	Low	0.586	0.23	41.4	-0.27
Vol Managed TSMOM	Null-Like / Inconclusive	Inconclusive	Low	0.561	0.25	43.9	-0.33
Rel Strength Mom	Likely Random	Luck	Moderate	0.882	0.05	11.8	-0.96
Random	Null-Like / Inconclusive	Inconclusive	Low	0.390	0.41	61.0	0.07

Interpretation:
Prom. = -log10(p), so larger values indicate rarer outcomes under the null.
Pctile and RCSI_z provide scale-free evidence and are more comparable across tickers.
Benchmark context appears in the comparison table and equity curve rather than this verdict summary.


QQQ: Regime-Conditional Skill Index by Strategy
RCSI (Actual Return − Median Monte Carlo Return, decimal)
1.20
0.800
0.400
0
-0.400
-0.800
-1.20
-0.322
Strongly
Random
-0.361
Strongly
Random
-0.034
Null-Like
Inconclusive
-0.129
Null-Like
Inconclusive
0.149
Null-Like
Inconclusive
Positive RCSI indicates performance above the simulated baseline (skill).
Negative RCSI suggests performance consistent with or below randomness.
Labels show the overall evidence class, not RCSI alone.
Likely Random
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy


QQQ: Trade-Level Return Ratio by Strategy and Market Regime
Gray hatched bars are suppressed because the regime cell has fewer than 20 trades.
Trend Pullback
Mean Rev Vol Filter
Rel Strength Mom
Breakout Vol+Mom
Vol Managed TSMOM
Random
0.300
0.150
0
-0.150
-0.300
-0.450
Trade-Level Return Ratio (unitless)
0.136
-0.211
-0.388
0.203
-0.012
0.074
-0.309
n=14
n=16
0.217
0.151
0.047
n=16
n=2
n=5
n=9
-0.050
0.143
Calm
Neutral
Stressed
Market Regime

QQQ: Strategy Performance Across Market Regimes

Gray cells are suppressed because the regime cell has fewer than 20 trades.

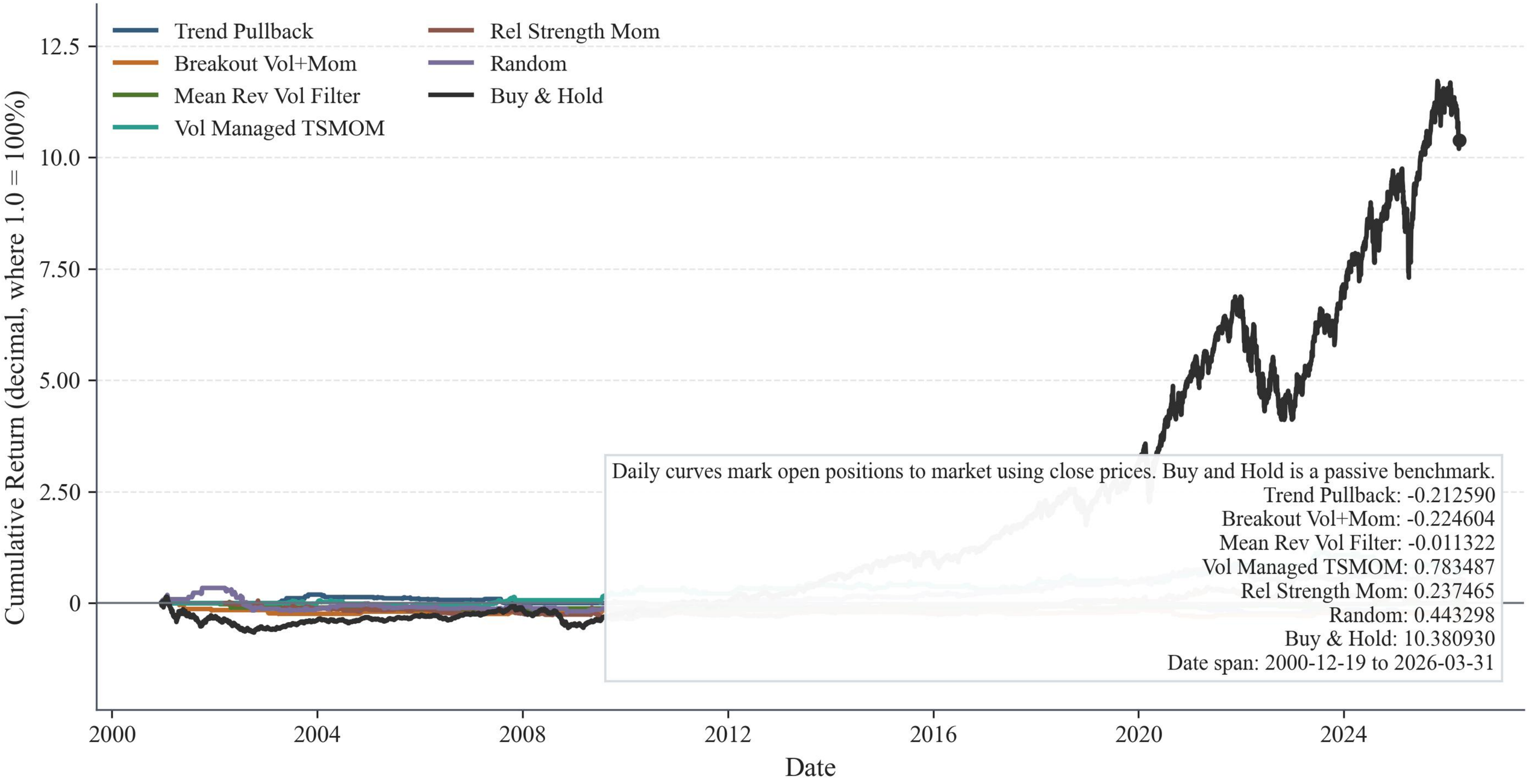
Strategy	Calm	Neutral	Stressed
Trend Pullback	0.14	-0.31	Suppressed n=16
Breakout Vol+Mom	-0.21	Suppressed n=14	Suppressed n=2
Mean Rev Vol Filter	-0.39	Suppressed n=16	Suppressed n=5
Vol Managed TSMOM	0.20	0.22	Suppressed n=9
Rel Strength Mom	-0.01	0.15	-0.05
Random	0.07	0.05	0.14

Market Regime

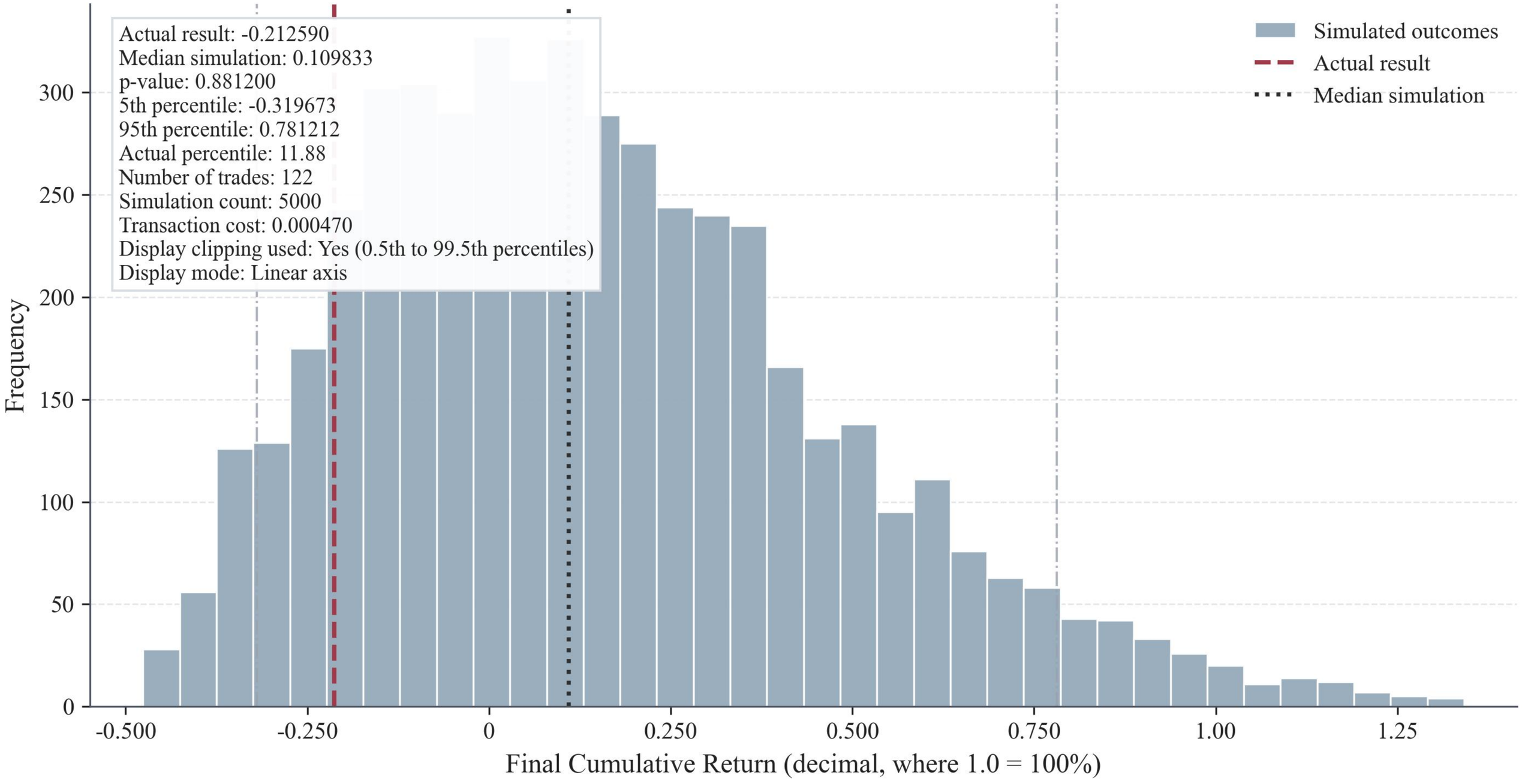
Trade-Level Return Ratio (unitless)

0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3

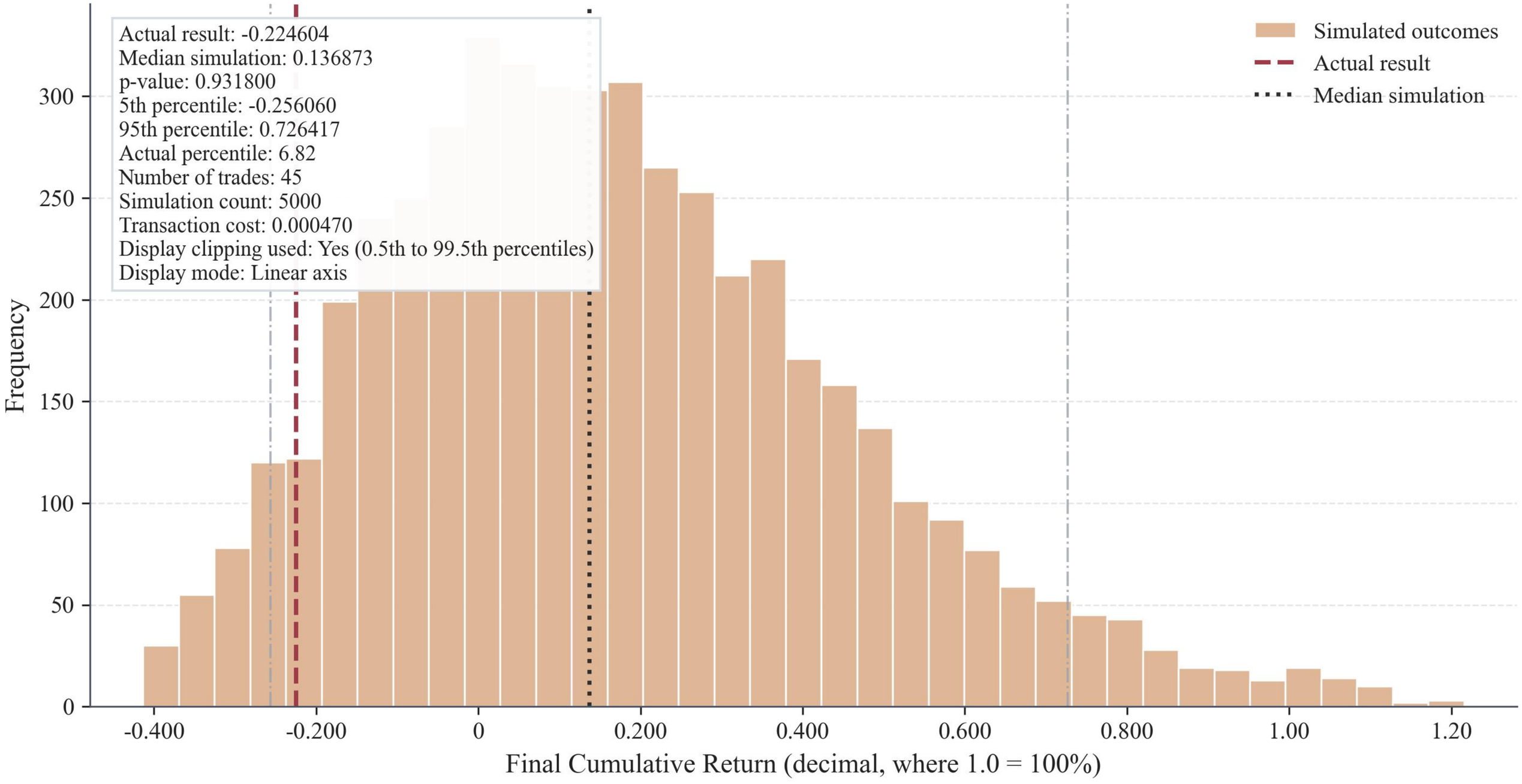
QQQ: Daily Equity Curves by Strategy and Benchmark



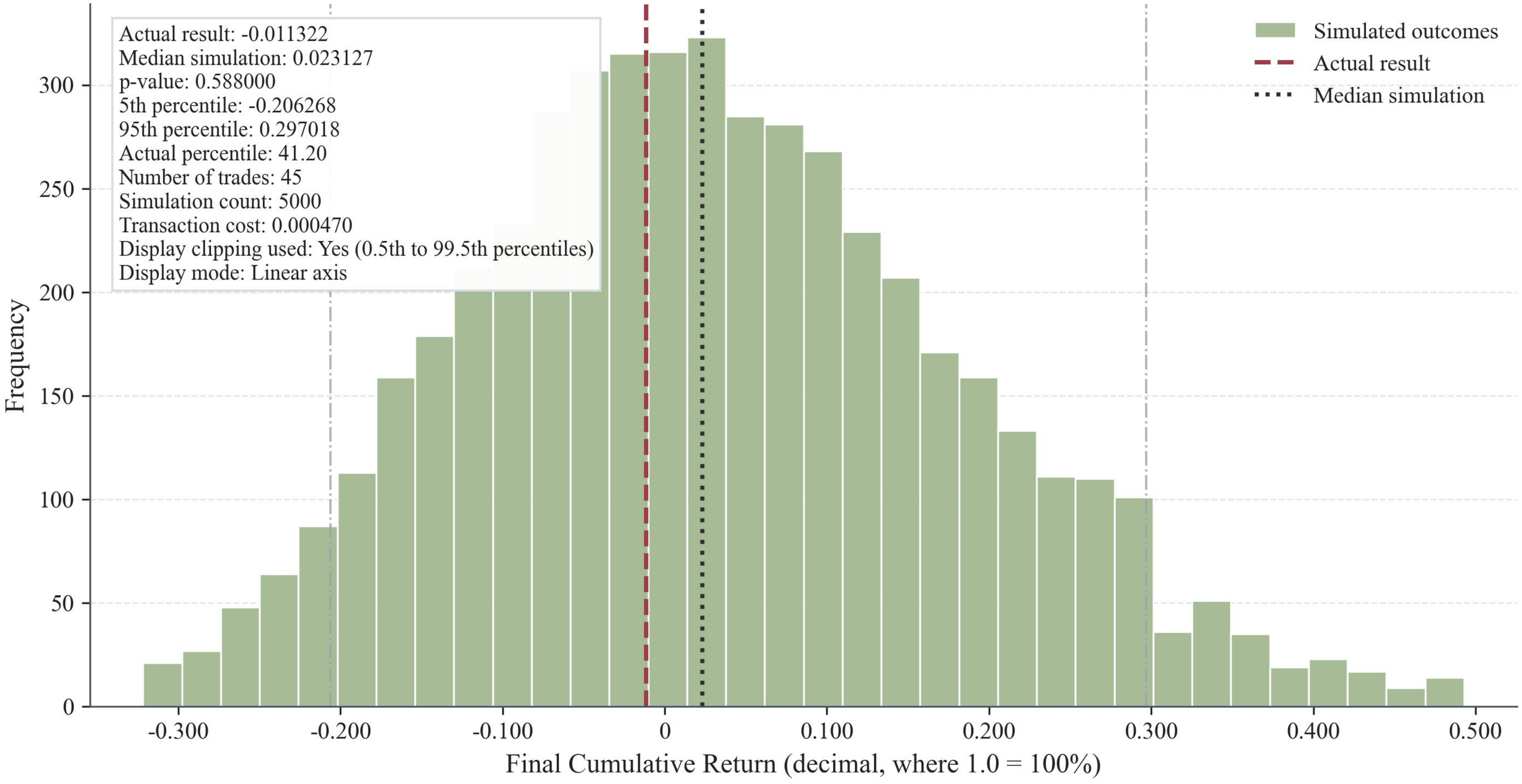
QQQ: Monte Carlo Distribution of Final Returns, Trend Pullback Strategy



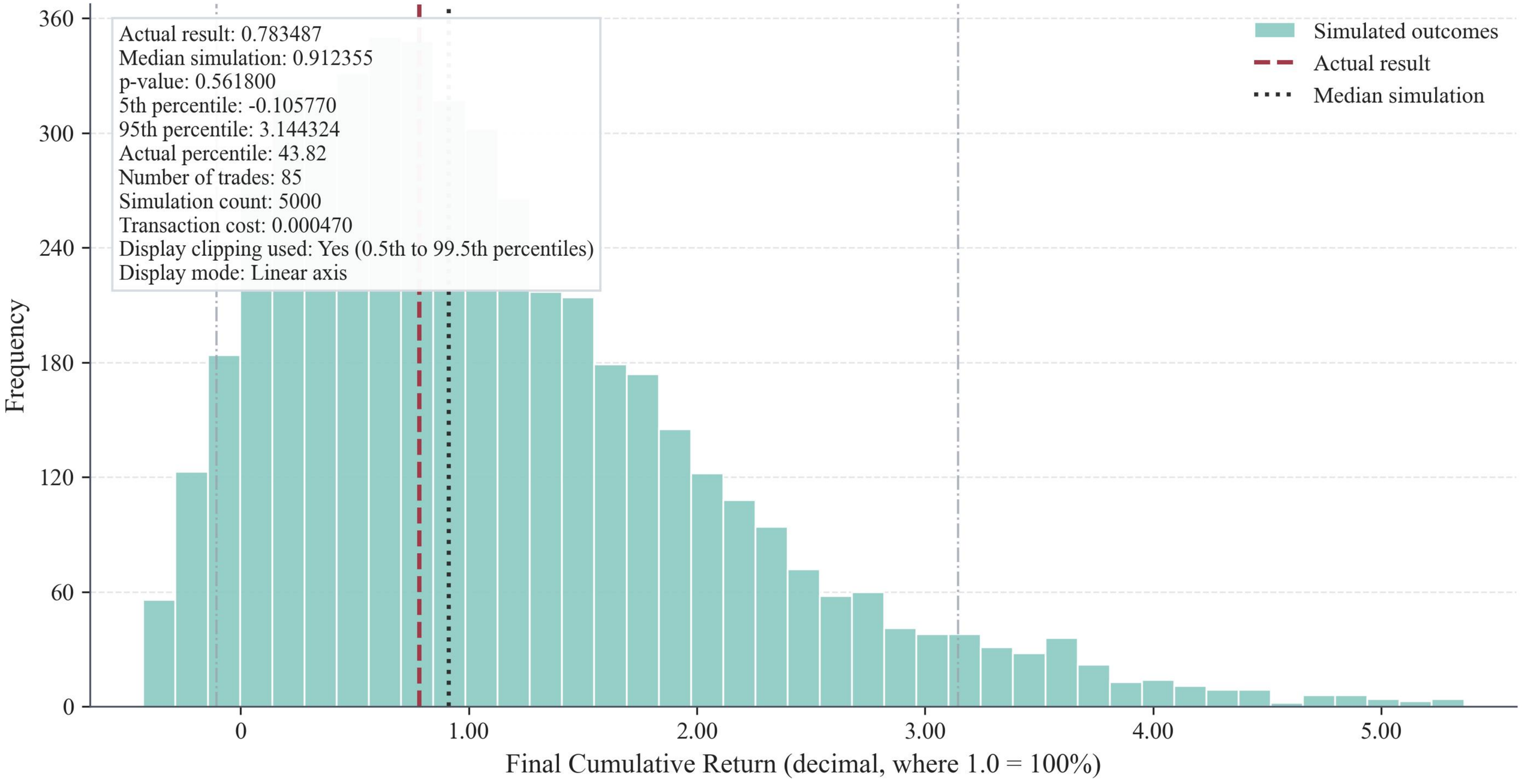
QQQ: Monte Carlo Distribution of Final Returns, Breakout Vol+Mom Strategy



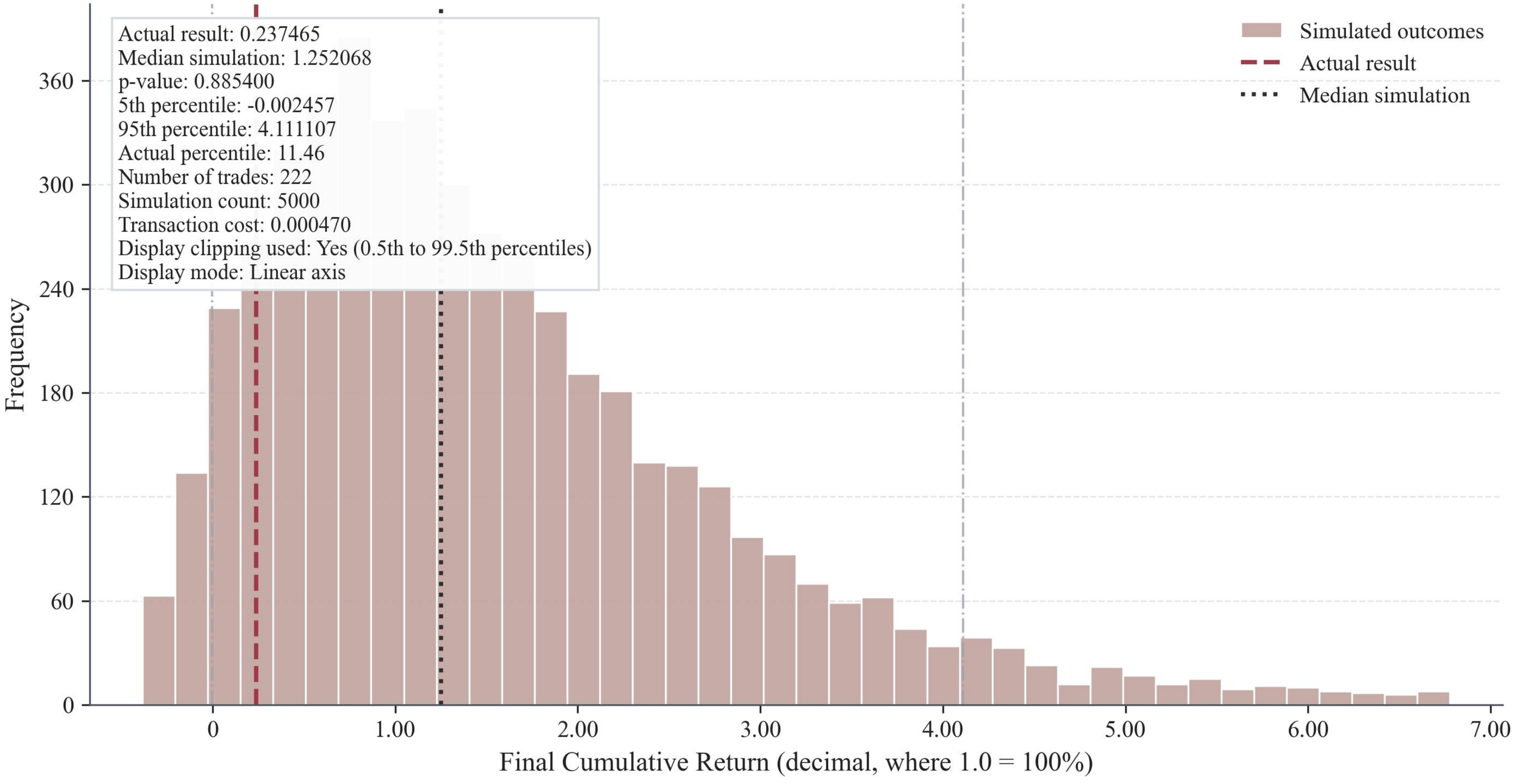
QQQ: Monte Carlo Distribution of Final Returns, Mean Rev Vol Filter Strategy



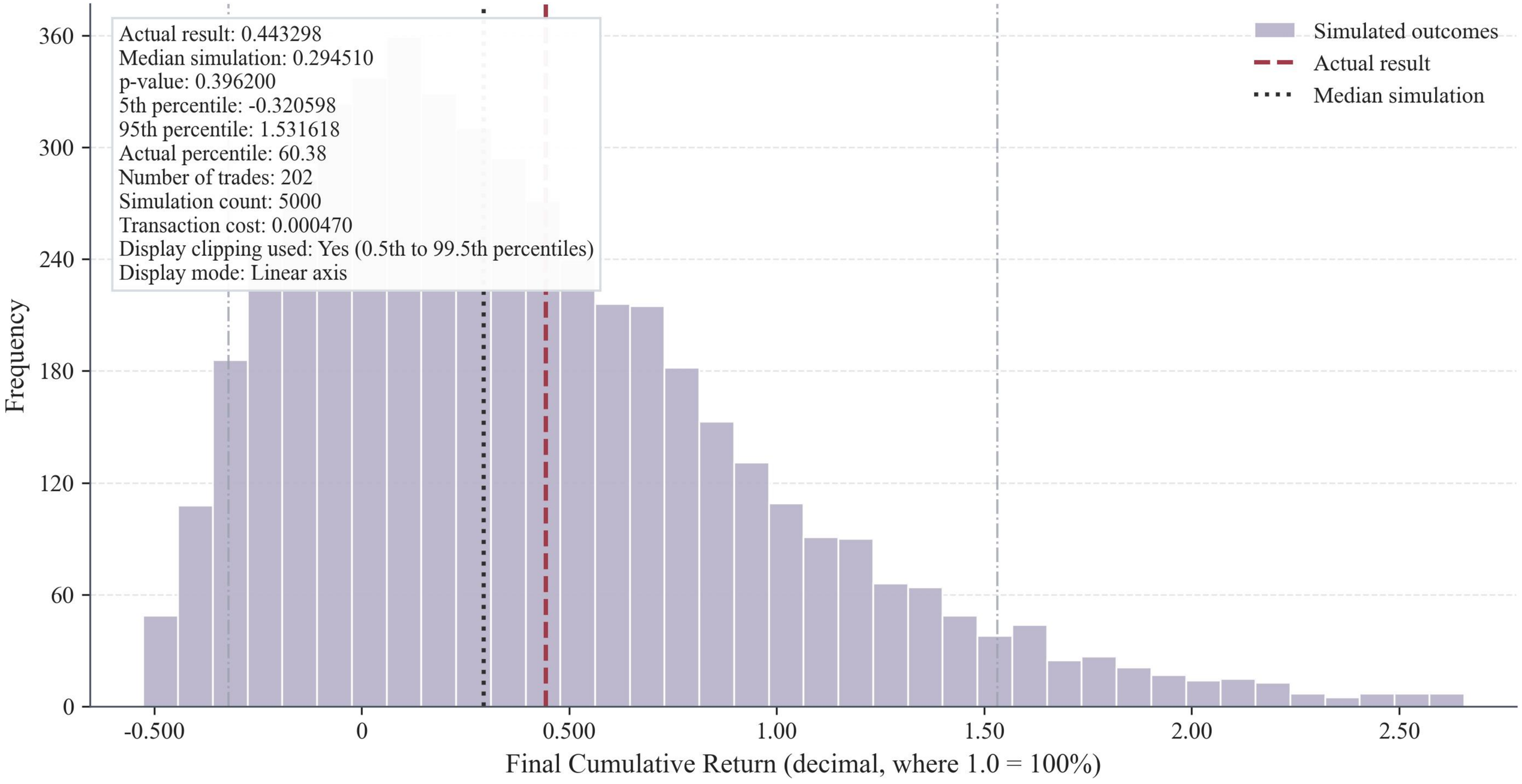
QQQ: Monte Carlo Distribution of Final Returns, Vol Managed TSMOM Strategy

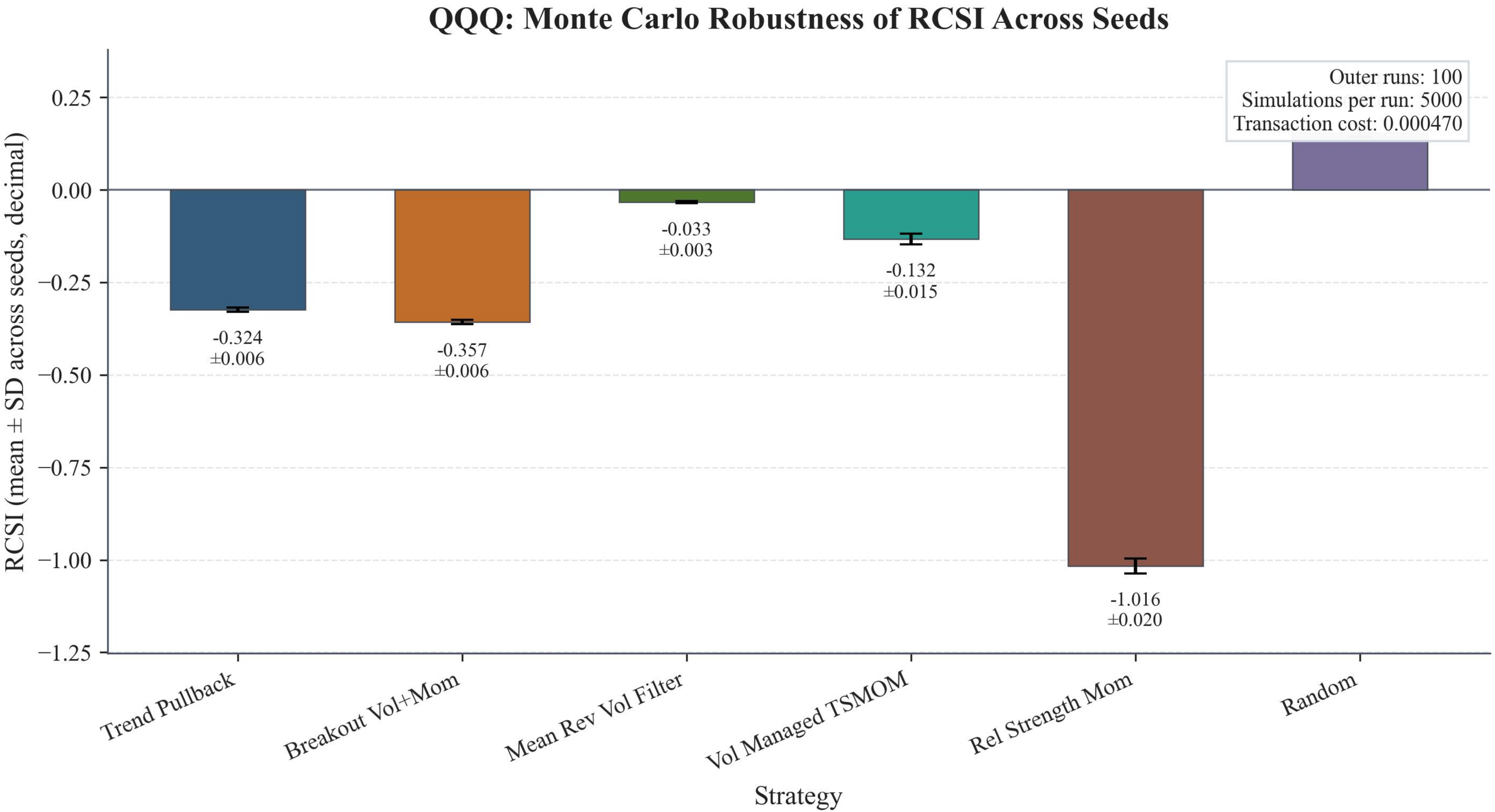


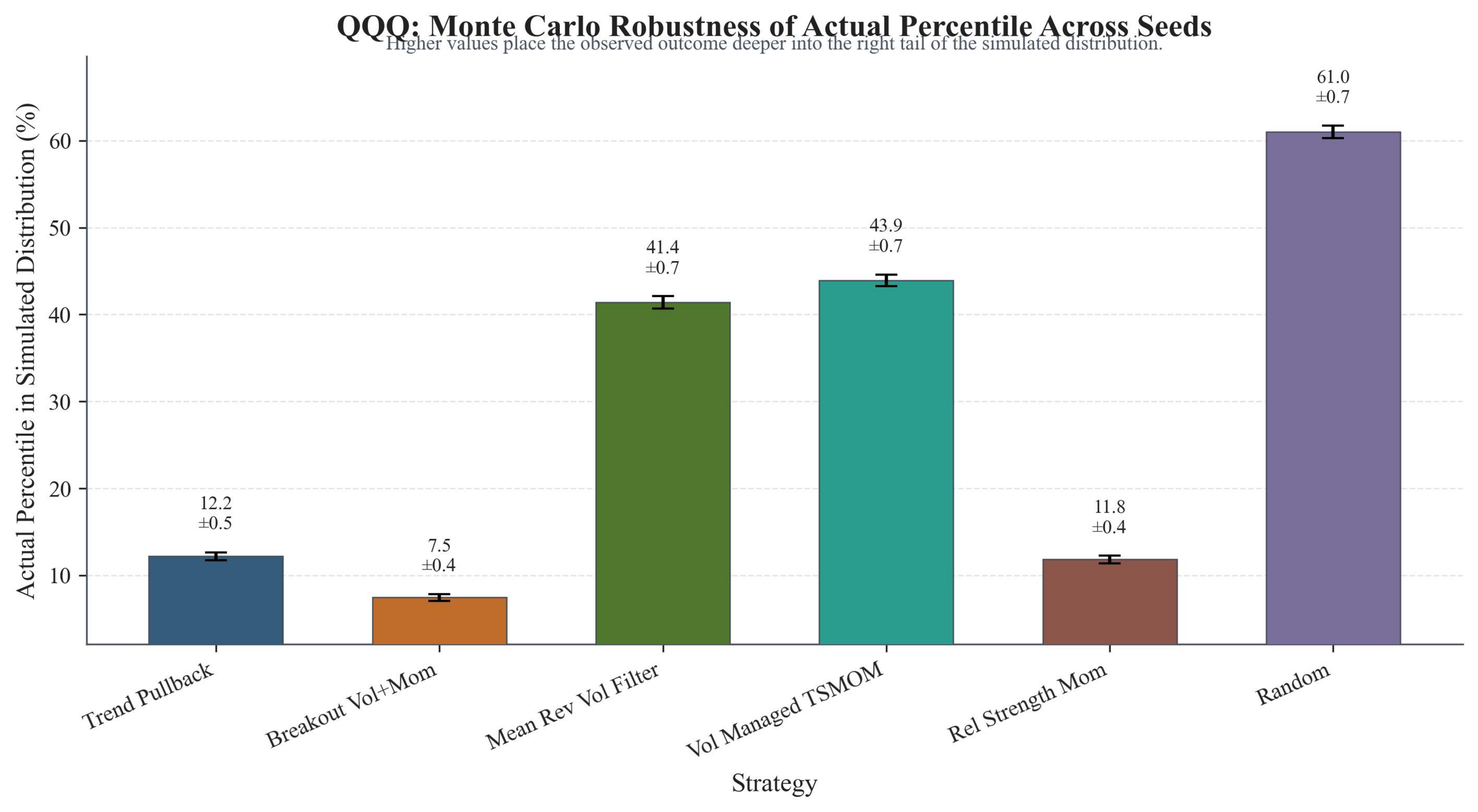
QQQ: Monte Carlo Distribution of Final Returns, Rel Strength Mom Strategy

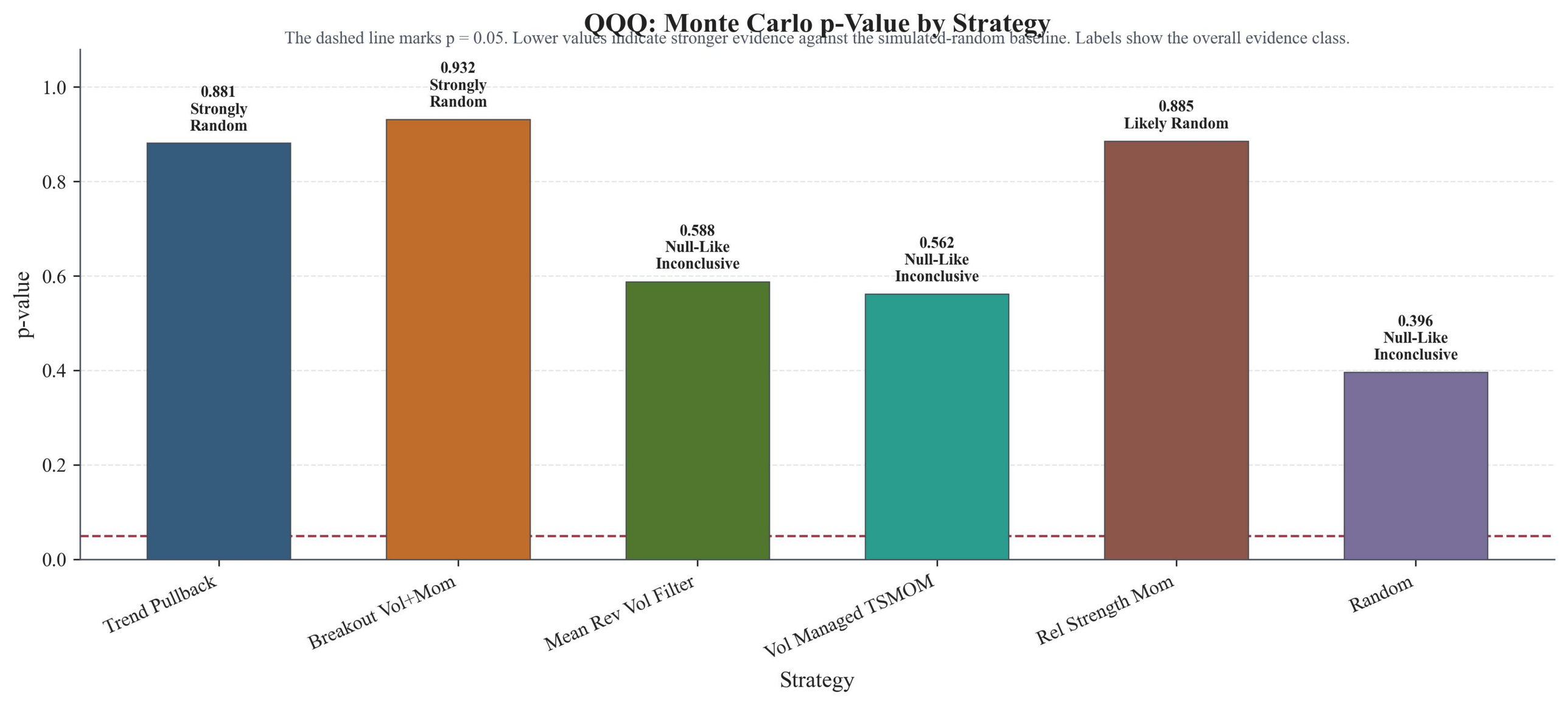


QQQ: Monte Carlo Distribution of Final Returns, Random Strategy




QQQ: Monte Carlo Robustness of RCSI Across Seeds
Outer runs: 100
Simulations per run: 5000
Transaction cost: 0.000470
RCSI (mean ± SD across seeds, decimal)
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
-0.324
±0.006
-0.357
±0.006
-0.033
±0.003
-0.132
±0.015
-1.016
±0.020
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy


QQQ: Monte Carlo Robustness of Actual Percentile Across Seeds
Higher values place the observed outcome deeper into the right tail of the simulated distribution.
Actual Percentile in Simulated Distribution (%)
60
50
40
30
20
10
12.2
±0.5
7.5
±0.4
41.4
±0.7
43.9
±0.7
11.8
±0.4
61.0
±0.7
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy


QQQ: Monte Carlo p-Value by Strategy
The dashed line marks p = 0.05. Lower values indicate stronger evidence against the simulated-random baseline. Labels show the overall evidence class.
p-value
1.0
0.8
0.6
0.4
0.2
0.0
0.881
Strongly
Random
0.932
Strongly
Random
0.588
Null-Like
Inconclusive
0.562
Null-Like
Inconclusive
0.885
Likely Random
0.396
Null-Like
Inconclusive
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy